Stage 6 — Check Your Understanding

Q1. Why the thickness drops out (Eq. 9.1)

The quality factor is q_F = –ln(P)/Δp. Substituting the attenuation and Darcy pressure-drop formulas:

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q_F = [4 α η L / (π d_f)] / [μ U L / k]

= 4 α η k / (π d_f μ U).
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Because a = d_f/2, k = (k/a²)a² = (k/a²)d_f²/4. Thus:

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q_F = α η (k/a²) d_f / (π μ U).
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Thickness L cancels because both attenuation –ln(P) and pressure drop Δp are proportional to L.

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q_F ≈ (0.1)(0.037)(1.25)(3×10■■) / [π(1.81×10■■)(0.1)]

≈ 2.44×10■³ Pa■¹.
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This agrees, to rounding of η and k/a², with the Stage 1 hexagonal Table 10 value 2.46×10■³ Pa■¹. The MPPS minimizes η because Δp and the other factors in q_F do not depend on particle diameter for a fixed design.

Q2. Proposition 10 with two numbers

In units of 10■³ Pa■¹: layer 1 has q■(0.2 μm)=3.0 and q■(0.4 μm)=4.0, so Q■=3.0 at 0.2 μm. Layer 2 has q■(0.2 μm)=3.5 and q■(0.4 μm)=2.0, so Q■=2.0 at 0.4 μm.

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q_stack(d; w) = w q■(d) + (1–w) q■(d).

w=0.5: q(0.2)=3.25; q(0.4)=3.00; Q=3.00.

q(0.2)=3.5–0.5w; q(0.4)=2+2w.

3.5–0.5w = 2+2w => w=0.6; Q=3.2.
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The optimized stack improves on layer 1 by (3.2–3.0)/3.0 = 6.67%. Condition (c) holds: q■(0.2)=3.5 > Q■=3.0. At one fixed size, a pressure-weighted average cannot exceed the better of its constituent layer values; stacking can only improve the minimum across different sizes.

Q3. Elasticities (Table 11)

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E_U = ln(0.858/6.73) / ln(0.2/0.05) = –1.486 ≈ –1.49.

q_F(2U)/q_F(U) ≈ 2^(–1.49) = 0.356.
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Doubling the face velocity, as when walking fast rather than resting, reduces q_F to about 36% of its previous value (a roughly 64% drop). The classical elasticity –1.39 gives 2^(–1.39)=0.382, or 38%; the practical advice is unchanged.

Q4. Why the base-design ratio changed (Section 9.2)

Stage 5 reported random/hexagonal q_F = 1.189 [1.128, 1.317], whereas Stage 6 reports 1.088 [1.017, 1.206]. The same physical cells and flow permeabilities k were used, so changing the sampled cell geometries cannot explain the difference. What changed were the stochastic particle draws and the particle-size grids used for the MPPS parabolic fit: Stage 5 used 0.12, 0.18, 0.25, 0.35 μm; Stage 6 used 0.13, 0.169, 0.22, 0.286 μm around the minimum.

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Hexagonal MPPS: 0.276 → 0.245 μm (–0.031 μm).
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Hexagonal q_F: 2.38 → 2.46 ×10■³ Pa■¹ (+3.36%).

Most of the ratio shift comes from the hexagonal estimate. The reported bootstrap intervals reflect particle and cell sampling but not the systematic effect of size-grid placement. The more robust conclusion across the designs is that for $\alpha \leq 0.1$, random identical-fibre arrays typically have a 9–24% higher q_F than hexagonal arrays, principally because of higher permeability.

Q5. Sensitivity and independence (Section 9.5)

The 32-fibre cells span two 16-fibre lengths with no mixing between halves. If a particle escaping the first half is also unusually likely to escape the second (for example, by remaining on an open streamline), then survival probabilities are positively correlated. Total attenuation $-\ln(1-f)$ grows less than proportionally with length, making the length-normalized single-fibre efficiency η lower in the long cells.

At 0.22 μm: η■■/η■■ = 1.043 [0.968, 1.124].

At 0.483 μm: η■■/η■■ = 1.081 [0.960, 1.218].

Neither interval shows the lower ratio expected from correlated escape; their lower bounds rule out a reduction exceeding about 3.2% and 4.0%, respectively, at the stated confidence level. The 0.483 μm test is stronger mechanistically because interception follows streamlines, so persistent open channels would be especially relevant. These results support the independent-layer capture assumption used in Proposition 10, within the resolution of this test; they do not prove perfect independence.